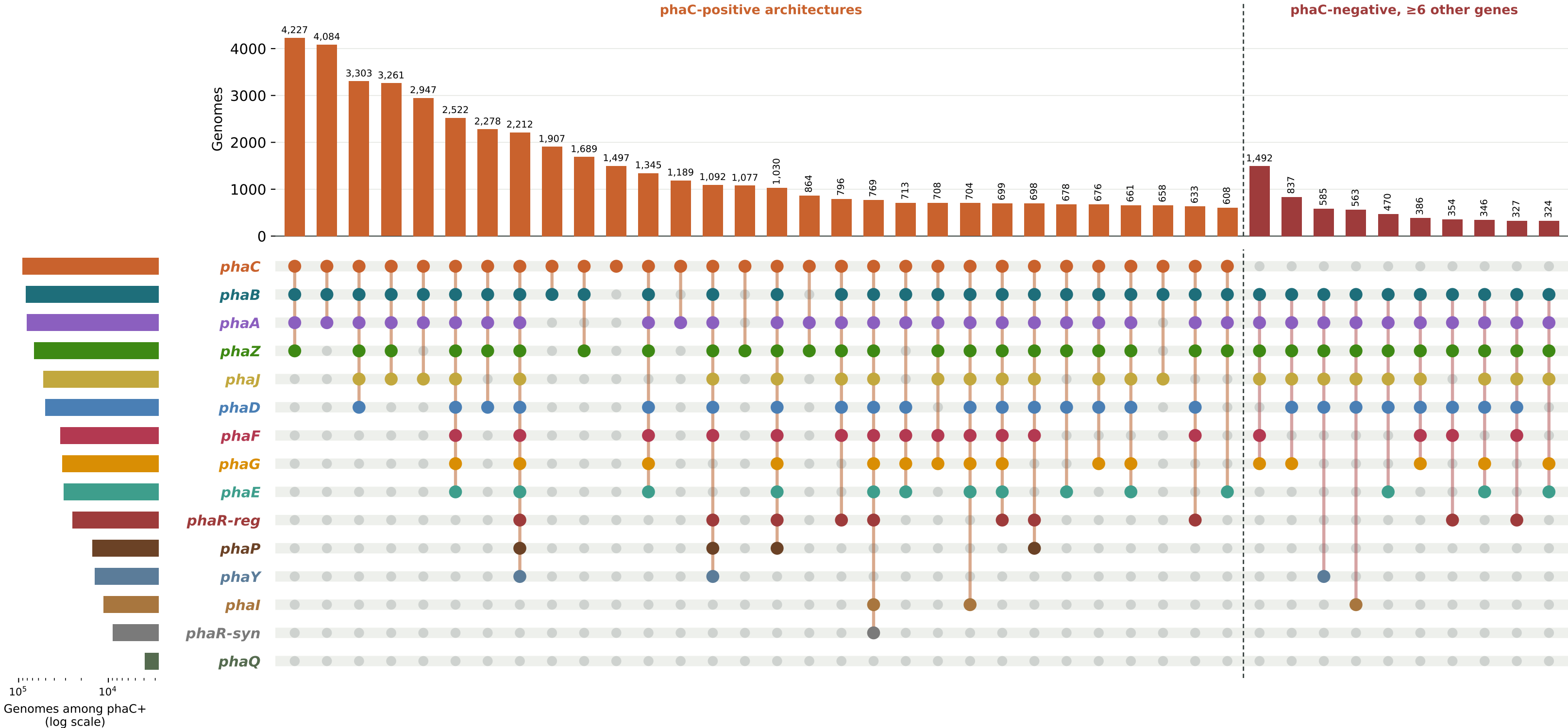


PHA Pathway Gene Architectures: Presence, Co-occurrence, and PhaC-Negative Outliers

Left: top 30 architectures among 90,846 phaC-positive genomes (1,779 distinct architectures total) | Right: top 10 of 686 architectures with phaC absent but >5 other PHA genes present



Architecture = the set of PHA pathway genes co-occurring in one genome (pipeline/pathway_architecture.py). Row order and left-panel totals are scoped to phaC-positive genomes only. Dot color = gene identity (matches row label and left bar); hollow/faint dot = absent. Right-hand columns (red) carry a substantial accessory-gene suite (phaA, phaB, and others) but no phaC hit -- either a genuinely divergent/incomplete pathway, or a phaC search miss on an otherwise PHA-gene-rich genome; worth targeted follow-up, not yet resolved.